Model Implementation Evidence

1. Objective

The objective of this stage was to implement the initial Pilot Assistance model in ODME using the requirements and operational concepts extracted from the source use case DOT/FAA/TC-21/48.

The implementation was performed incrementally so that the relationship between model structure, metadata, scenario derivation, and generated artifacts could be observed.

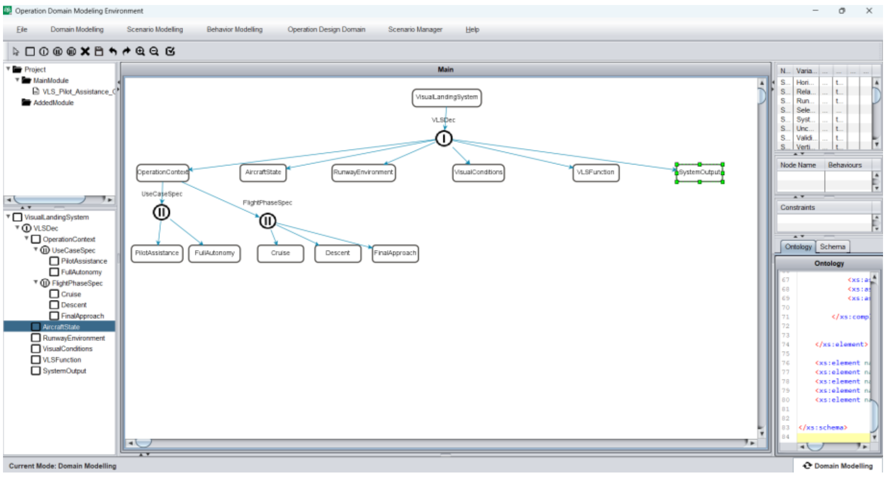
2. Domain Modelling Procedure

The following workflow was performed:

1. A new ODME project was created.
2. The root entity `VisualLandingSystem` was established.
3. The main aspect structure was created.
4. Operational entities were added.
5. Use case and flight phase specializations were added.
6. Variables were added to the relevant entities.
7. Behaviours were added to `VLSFunction`.
8. The domain model was saved.
9. The generated schema representation was inspected.

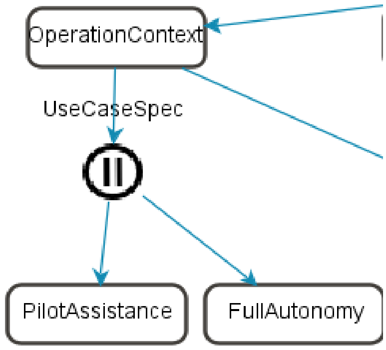
3. Model Evidence

Figure 1. Complete Pilot Assistance domain model



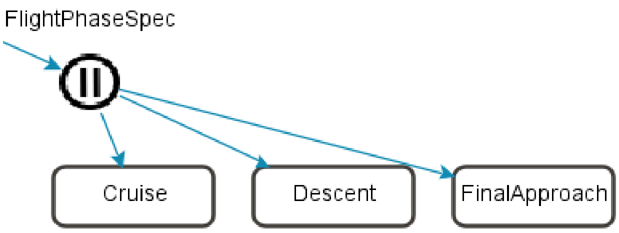
Caption:
Initial domain model developed for the Pilot Assistance use case. The model separates operational context, aircraft state, runway conditions, visual conditions, VLS functions, and system outputs.

Figure 2. UseCaseSpec specialization



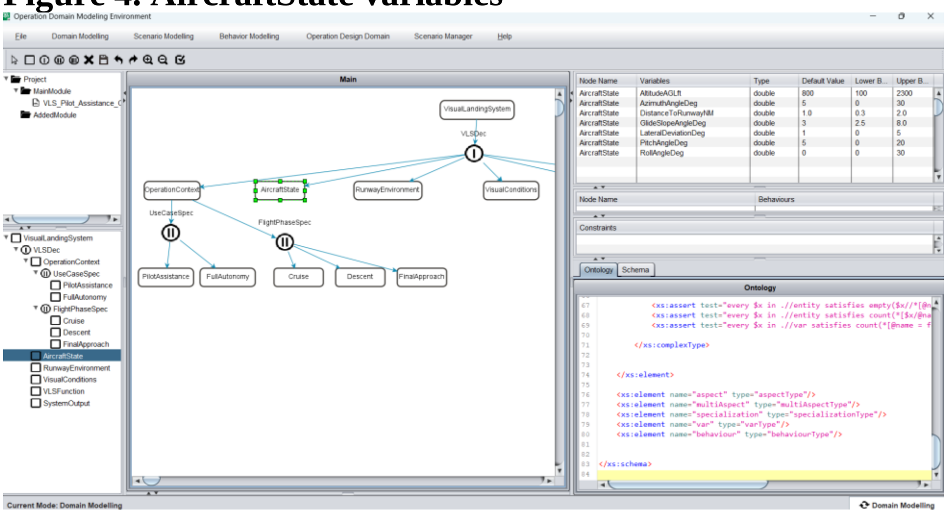
Caption:
Use case specialization containing PilotAssistance and FullAutonomy alternatives. PilotAssistance was selected during the first scenario derivation.

Figure 3. FlightPhaseSpec specialization



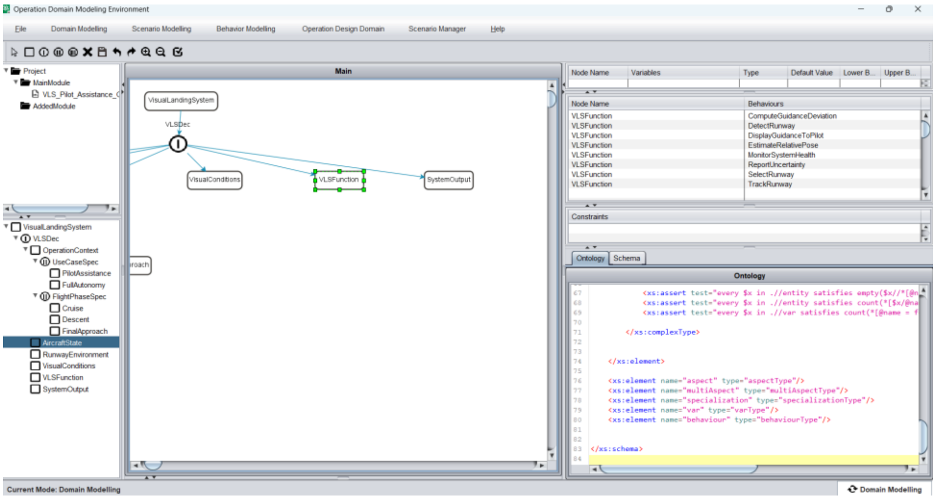
Caption:
Flight phase specialization used to derive scenario-specific operational configurations.

Figure 4. AircraftState variables



Caption: Aircraft state parameters implemented from the documented operating limits. Scenario values were selected within these limits for workflow testing.

Figure 5. VLSFunction behaviours



Caption:
Initial system-level functional behaviours implemented for the Pilot Assistance workflow.

4. Implementation Result

The initial domain model was successfully created and saved. ODME maintained the model structure and associated metadata, allowing the project to proceed to Scenario Modelling.